

// FIELD NOTES • NO. 01 • INTERSTELLAR

Two probes, forty-eight years, *still talking.*

Launched in 1977 to tour the outer planets, Voyager 1 and 2 kept going. They are now the most distant human-made objects in existence — small, cold, plutonium-warmed machines whispering back across light-hours of empty space.

// 01 — STATUS

APPROXIMATE • EARLY 2026

V1 DISTANCE

~167

AU from the Sun
— about 23 light-
hours

CRUISE SPEED

~17

km/s, V1 relative
to the Sun

MISSION AGE

48

years since
launch, 1977 →
now

POWER LEFT

~30%

RTG output,
fading ~4W per
year

// 02 — TRAJECTORY

DISTANCE FROM THE SUN

Falling outward, *forever*.

Neither probe will ever return. With no atmosphere to slow them and no destination to reach, they coast outward at a steady clip — Voyager 1 a little faster, having taken the more aggressive slingshot past Saturn.



— Voyager 1 — Voyager 2

// 03 — THE GRAND TOUR

A RARE ALIGNMENT

A planetary alignment that *comes once in 176 years*.

The mission existed because the outer planets happened to line up so a single craft could swing from one to the next, each gravity assist flinging it onward. Miss the window and you'd wait until the 2150s.

- **1977** Both launch. Voyager 2 leaves first, on Aug 20; Voyager 1 follows on Sep 5, on a faster track.
- **1979** **Jupiter.** Active volcanoes found on the moon Io — the first ever seen beyond Earth.
- **1980-81** **Saturn.** Voyager 1 dives past Titan, bending its path up and out of the planetary plane for good.
- **1986** **Uranus.** Voyager 2 alone, the only craft ever to visit.
- **1989** **Neptune.** The last planet. Voyager 2 completes the tour and turns toward deep space.

- 1990 The Pale Blue Dot. Voyager 1 turns back for a final family portrait of the solar system.
- 2012 Interstellar. Voyager 1 crosses the heliopause — the edge of the Sun's bubble.
- 2018 Voyager 2 follows across the heliopause, six years behind its twin.

// 04 — THE CRAFT

TWINS, NOT IDENTICAL

Less computing power than *a key fob*.

Each probe carries about 70 kilobytes of memory and is warmed by the slow decay of plutonium-238. Engineers now switch instruments off one at a time to stretch the dwindling power budget for a few more years.

PARAMETER	VOYAGER 1	VOYAGER 2
Launched	Sep 5, 1977	Aug 20, 1977
Planets visited	2	4
Crossed heliopause	2012	2018
Speed (km/s)	~17.0	~15.4
Distance (AU, approx)	~167	~142
One-way signal time	~23 hr	~19 hr

A

The big dish

A 3.7-metre high-gain antenna stays locked on Earth, sending data home at a trickle — a few hundred bits per second.

22 W transmitter

B

Nuclear warmth

Three radioisotope generators turn the heat of decaying plutonium into electricity. No sunlight reaches this far out.

-4 W / year

C

Ancient memory

Commands run on 1970s computers with roughly 70 KB of storage — millions of times less than the phone in your pocket.

~70 KB

A gold-plated *hello*, for whoever finds it.

Bolted to each probe is a phonograph record of copper plated in gold, etched with instructions for playback. It is built to survive a billion years in the vacuum — a portrait of Earth assembled in case anyone, ever, is listening.

FORMAT • 01

12-inch disc

Gold-plated copper, with a cartridge and stylus included. A diagram on the cover shows how to play it.

GREETINGS • 02

55 languages

Spoken hellos from Akkadian to Wu, plus a message from the UN Secretary-General.

IMAGES • 03

115 photos

Encoded as audio: anatomy, landscapes, a supermarket, a nursing mother, the structure of DNA.

MUSIC • 04

90 minutes

Bach and Chuck Berry, Javanese gamelan, a Navajo night chant, a Bulgarian shepherdess.

This is a present from a small, distant world, a token of our sounds, our science, our images, our music, our thoughts and our feelings.

— CARTER'S MESSAGE, ETCHED ABOARD • 1977

// Field note: figures here are approximate and rounded for legibility. Live telemetry — exact distance, speed, and remaining power — drifts daily and is best taken from NASA's Voyager mission status page rather than this document. I don't have live access and some current numbers may be out of date.

SIGNAL.LOCK // OK

vgr.